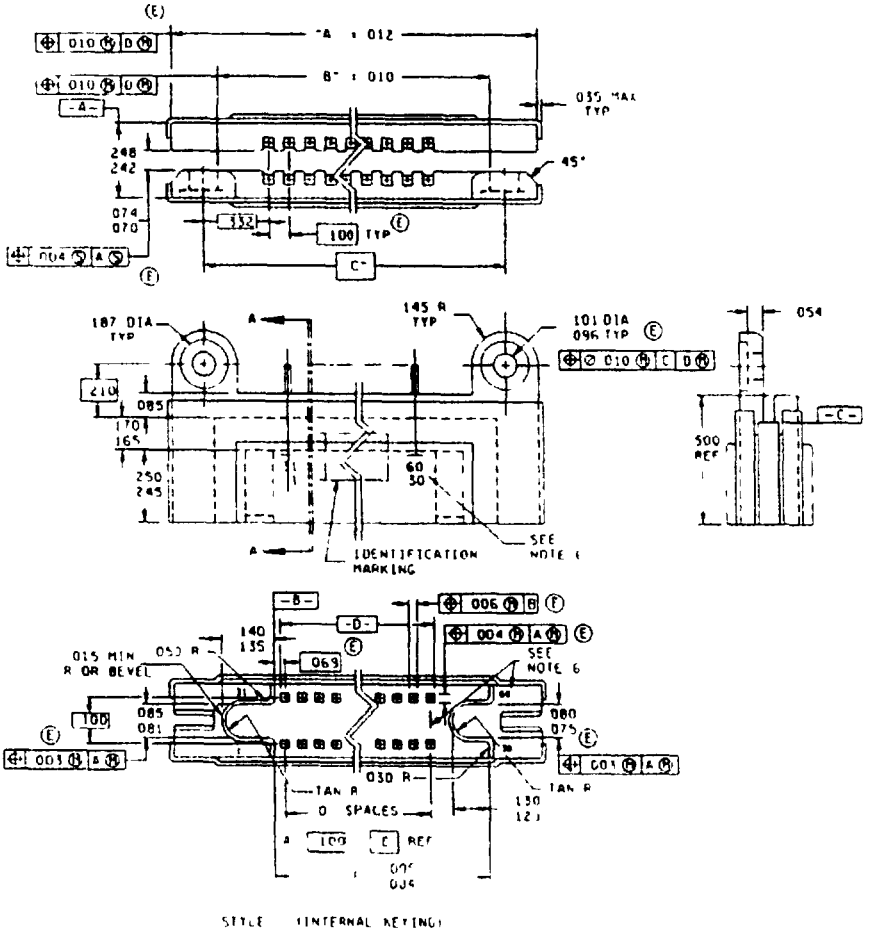
MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES PLUG, 30 THRU 140 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SQ GRID)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-55302



(E) FIGURE 1 Connectors, plug, 100 (2.54 mm, square grid

(F) denotes changes

DISTRIBUTION STATEMENT A Approved for public release; distribution is unlimited.

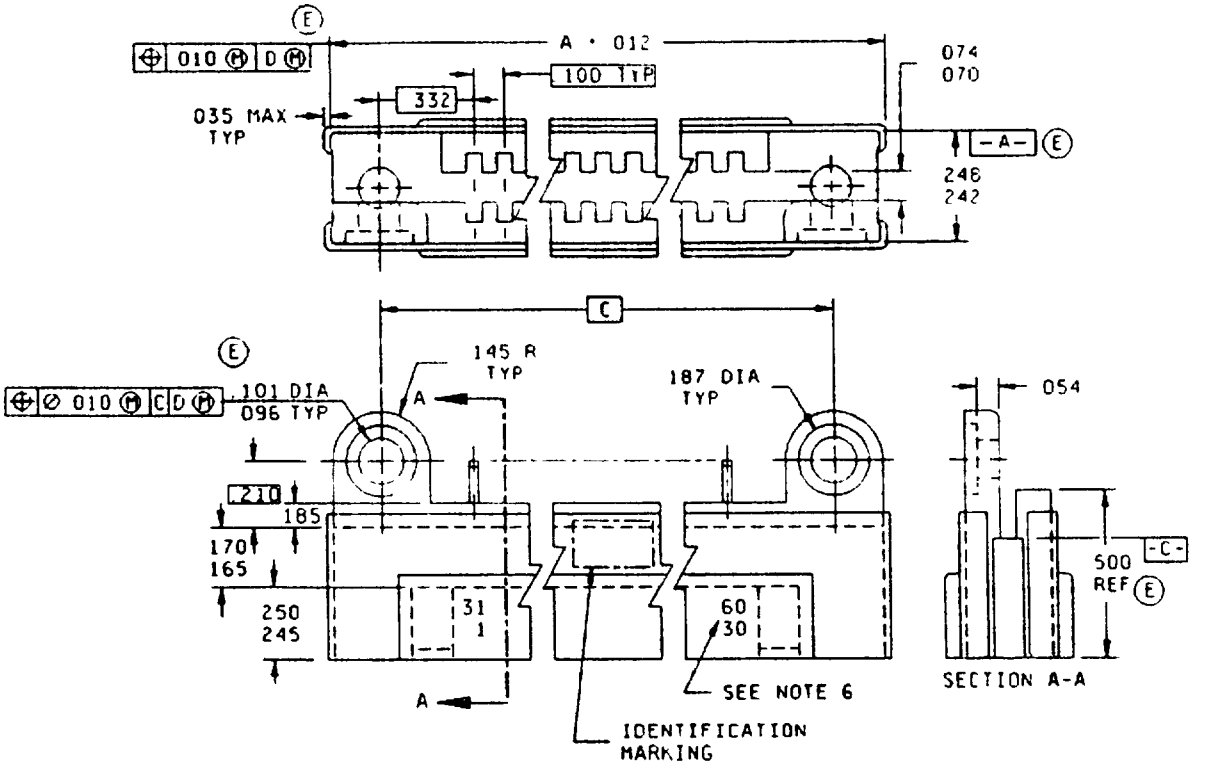
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:45Z
Check the source to verify that this is the current version before use.

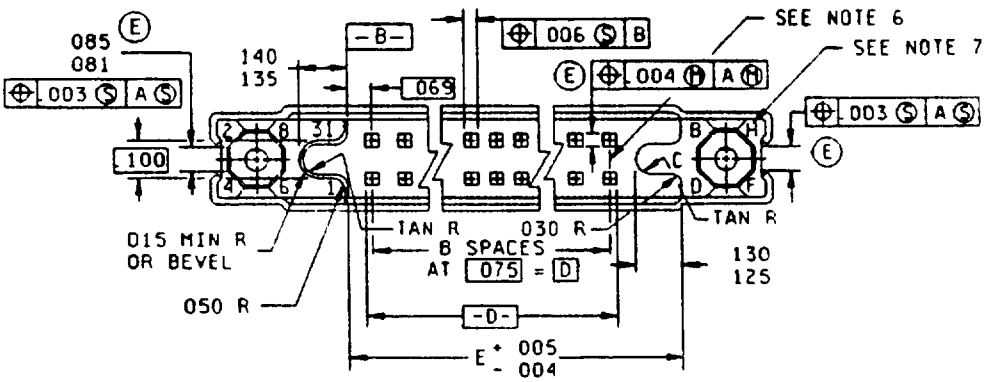
Dash 1/ numbers	Number of contacts	Dimensions						Contact identification numbers
		A	B	C	D	E	F	
01	50	3 355 (85 22)	2 955 (75 06)	24	3 065 (77 85)	2 400 (60 96)	2 675 (67 95)	1 - 25 26 - 50
02	60	3 855 (97 92)	3 455 (87.76)	29	3 565 (90.55)	2 900 (73.66)	3 175 (80.65)	1 - 30 31 - 60
03	80	4 855 (123 32)	4 455 (113 16)	39	4.565 (115 95)	3 900 (99 06)	4 175 (106 05)	1 - 40 41 - 80
04	90	5 355 (136 02)	4 955 (125 86)	44	5 065 (128 65)	4 400 (111 76)	4 675 (118 75)	1 - 45 46 - 90
05	100	5 855 (148 72)	5 455 (138 56)	49	5 365 (141 35)	4 900 (124 46)	5 175 (131 45)	1 - 50 51 - 100
06	110	6 355 (161 42)	5 955 (151 26)	54	6 065 (154 05)	5 400 (137 16)	5 675 (144 15)	1 - 55 56 - 110
07	30	2 355 (59 82)	1 955 (49 66)	14	2 065 (52 45)	1 400 (35 56)	1 675 (42 55)	1 - 15 16 - 30
08	40	2 855 (72 52)	2 455 (62 36)	19	2 565 (65 15)	1 900 (48 26)	2 175 (55 25)	1 - 20 21 - 40
09	70	4 355 (110 62)	3 955 (100 46)	34	4.065 (103 25)	3 400 (86 36)	3.675 (93 35)	1 - 35 36 - 70

1/ Dash number 01 through 9 are for style 1 (internal keying) connectors

(E) FIGURE 1 Connectors, plug, 100 (2 54 mm) square grid - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:45Z
Check the source to verify that this is the current version before use.





STYLE 2 (EXTERNAL KEYING)

E FIGURE 1 Connectors, plug, 100 (2.54 mm) square grid - Continued

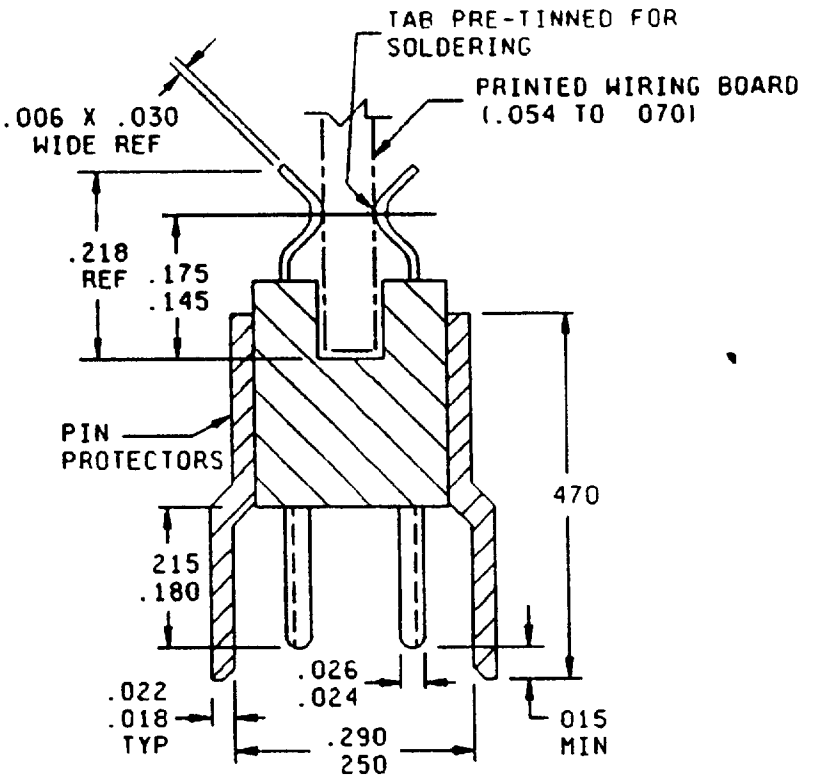
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:45Z
Check the source to verify that this is the current version before use.

Dash 1/ numbers	Number of contacts	Dimensions					Contact identification numbers
		A	B	C	D	E	
10	30	2 355 (59 82)	14	2 065 (52 45)	1 400 (35 56)	1 675 (42 55)	1 - 15 16 - 30
11	40	2 855 (72 52)	19	2 656 (65 15)	1 900 (48.26)	2.175 (55.25)	1 - 20 21 - 40
12	50	3 355 (85 22)	24	3 065 (77 85)	2 400 (60 96)	2 675 (67.95)	1 - 25 26 - 50
13	60	3 855 (97 92)	29	3 565 (90 55)	2 900 (73 66)	3.175 (80 65)	1 - 30 31 - 60
14	70	4 355 (110 62)	34	4 065 (103 25)	3 400 (86 36)	3 675 (93 35)	1 - 35 36 - 70
15	80	4 855 (123 32)	39	4 656 (111 95)	3 900 (99 06)	4 175 (106 05)	1 - 40 41 - 80
16	90	5 355 (136 02)	44	5 065 (128 65)	4 400 (111 76)	4 675 (118 75)	1 - 45 46 - 90
17	100	5 855 (148 72)	49	5 565 (141 35)	4 900 (124 46)	5 175 (131 45)	1 - 50 51 - 100
18	110	6 355 (161 42)	54	6 065 (154 05)	5 400 (137 16)	5 675 (144 15)	1 - 55 56 - 110
19	120	6 855 (174 18)	59	6 565 (166 75)	5 900 (149 86)	6 175 (156 85)	1 - 60 61 - 120
20	130	7 355 (186 82)	64	7 065 (179 45)	6 400 (162 56)	6 675 (169 55)	1 - 65 66 - 120
21	134	7 555 (191 90)	66	7 265 (184 53)	6 600 (167 64)	6 875 (174 63)	1 - 67 68 - 134
22	140	7 855 (199 52)	69	7 565 (192 15)	6 900 (175 26)	7 175 (182.25)	1 - 70 71 - 140

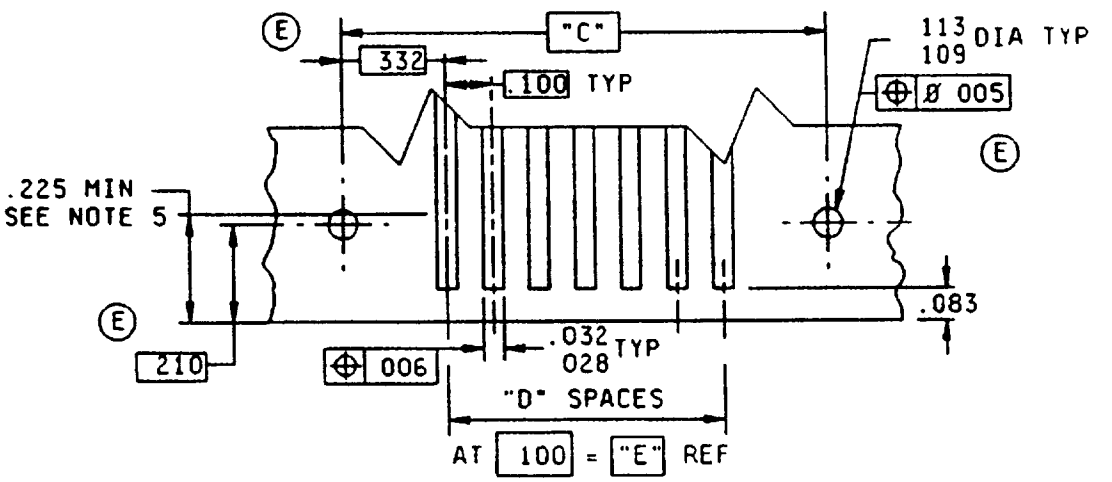
1/ Dash number 10 through 22 are for style 2 (external keying) connectors

(E) FIGURE 1 Connectors, plug, 100 (2.54 mm) square grid Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:45Z
Check the source to verify that this is the current version before use.



SECTION A-A



RECOMMENDED PRINTED WIRING BOARD LAYOUT

(E) FIGURE 1 Connectors, plug, 100 (2.54 mm) square grid - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:45Z
Check the source to verify that this is the current version before use.

Inches	mm	Inches	mm	Inches	mm
003	0 08	070	1 78	170	4 32
004	0 10	074	1.88	175	4 44
005	0 13	075	1 90	180	4 57
006	0 15	080	2.03	187	4 75
008	0 20	081	2 06	208	5 28
010	0 25	083	2 11	210	5 33
012	0.30	085	2 16	212	5.38
015	0 38	096	2 44	215	5 46
018	0 46	100	2 54	218	5 54
022	0 56	101	2 57	225	5 72
024	0 61	109	2 77	242	6 15
026	0 66	113	2 87	245	6 22
028	0 71	125	3 18	248	6 30
030	0 76	130	3 30	250	6 35
032	0 81	135	3 43	290	7 37
035	0 89	140	3 56	332	8 43
050	1 27	145	3 68	470	11 94
054	1 37	165	4 19	500	12 70
069	1 75				

NOTES

1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ± 005 (0.13 mm) on decimals and ±2° on angles
4. These connectors mate with connectors specified in MIL-C-55302/27
5. Minimum pad length on printed circuit board
6. Numbers indicating, every 5 cavities marked or molded on side. Numbers indicating end cavities and lines indicating every 5 cavities molded or marked on mating face
7. Location indicators embossed on surface

(E) FIGURE 1. Connectors, plug, .100 (2.54 mm) square grid - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:45Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

Design and construction

Dimensions and configuration See figure 1

Material

(E) Insulator body Insulator material shall be in accordance with MIL-C-55302 or MIL-M-24519 type GLCP-30r

(E) Contact material Contact material shall be in accordance with MIL-C-55302 or beryllium copper alloy CDA174 in accordance with ASTM B768

Plating The contact plating for the engagement area, 150 inch minimum length, shall be gold per MIL-C-55302 over nickel per MIL-C-55302 The contact plating for the solder tail area, 160 inch minimum length, shall be tin lead over nickel, with tin lead dip per MIL-C-55302 for 080 minimum length The remainder of the contact shall be nickel plated per MIL-C-55302

Contact identification See figure 1

Coating The aluminum pin protector shall be coated per MIL-A-8625, type II or epoxy coated 001 to 003 inch thick

Contact rating 3 0 amperes maximum per contact, 2 25 amperes continuous per contact at 75°F

Keying See MIL-C-55302/31

Mating and unmating The maximum mating force in pounds shall be the numberof contacts multiplied by 25 and the withdrawal force in pounds shall be a minimum of 025 times the number of contacts and shall not exceed the measured insertion force

Contact resistance The average contact resistance of all contacts measured shall not exceed 015 ohm, and no individual contact pair shall have a resistance exceeding 020 ohm

Contact retention 3 pounds minimum

Dielectric withstanding voltage

Sea level 900 volts rms

High altitude 200 volts rms

Part number M55302/26- (dash number from figure 1)

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:45Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians
Army - CR
Navy - EC
Air Force - 17

Review activities
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 99
DLA - ES

User activities
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity
Army - CR

Agent
DLA - ES

(Project 5935-3907-01)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:45Z
Check the source to verify that this is the current version before use.